

Deterministic Optimization

Linear Programming
Duality

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Table of Possibles and Impossibles

LP Duality

Learning Objectives for this lesson

- Further explore implications of weak and strong duality

Tables of Possibles & Impossibles

	Finite Optimal	Unbounded Optimal	Infeasible
Finite Optimal	Possible/Impossible?	Possible/Impossible?	Possible/Impossible ?
Unbounded Optimal	Possible/Impossible?	Possible/Impossible?	Possible/Impossible ?
Infeasible	Possible/Impossible?	Possible/Impossible?	Possible/Impossible ?

Figuring Out the Table

1. Primal *finite optimal*, Dual *finite optimal*: **Possible**. In this case, the strong duality tells us that $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$ for primal optimal solution $\mathbf{x}^*$ and dual optimum $\mathbf{y}^*$. The strong duality actually tells us: if Primal *finite optimal*, Dual **must** be *finite optimal*, and vice versa.
2. Primal *finite optimal*, Dual *unbounded*: **Impossible** due to the weak duality. If the dual has unbounded optimum, namely $+\infty$, it can not be a lower bound for the finite optimal cost of the primal.
3. Primal *finite optimal*, Dual *infeasible*: **Impossible** due to the strong duality, which says if the primal has a finite optimal cost, so does the dual.
4. Primal *unbounded*, Dual *finite optimal*: **Impossible** due to the weak duality. It can also be seen by symmetry with the case Primal *finite optimal*, Dual *unbounded*.

Figuring Out the Table (cont.)

5. Primal *unbounded*, Dual *unbounded*: **Impossible** due to the weak duality. Both unbounded means that Primal optimal cost is $-\infty$, and dual optimal cost is $+\infty$, which can never happen.
6. Primal *unbounded*, Dual *infeasible*: **Possible** due to the weak duality. Primal has $-\infty$ cost, the dual cannot have any feasible solution, otherwise that feasible solution would give a lower bound to $-\infty$ by weak duality, which is clearly impossible. The weak duality actually tells us: if Primal *unbounded*, Dual **must** be *infeasible*.
7. Primal *infeasible*, Dual *finite optimal*: **Impossible** by using the result of Primal *finite optimal*, Dual *infeasible*.
8. Primal *infeasible*, Dual *unbounded*: **Possible** by using the result of Primal *unbounded*, Dual *infeasible*. But notice that when Primal *infeasible*, the dual may not necessarily be *unbounded*. In other words, it is not true that if Primal *infeasible*, the dual must be *unbounded*. The reason is exactly the next situation.

Figuring Out the Table (cont.)

9. Primal *infeasible*, Dual *infeasible*: **Possible**. Not directly covered by weak duality or strong duality. But look at the following example.

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 = 1 \\ & 2x_1 + 2x_2 = 3 \\ & x_1, x_2 \text{ free.}\end{array}$$

$$\begin{array}{ll}\max & y_1 + 3y_2 \\ \text{s.t.} & y_1 + 2y_2 = 1 \\ & y_1 + 2y_2 = 2 \\ & y_1, y_2 \text{ free.}\end{array}$$

	Finite Optimal	Unbounded Optimal	Infeasible
Finite Optimal	Possible	Impossible	Impossible
Unbounded Optimal	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

Summary

- We learned:
 - All possible and impossible cases of primal and dual solutions using weak and strong duality